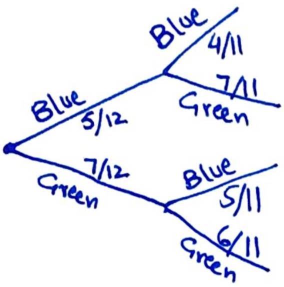
QUESTION # 1

10/10

Blue balls : 5   Gray Balls : 7   Ligma Balls : 12

i)



$\frac{20}{132} = \frac{10}{66} = \frac{5}{33}$ (BB, both are blue) 4

$\frac{35}{132}$ (BG) or {B, G} (first blue, second green)

$\frac{35}{132}$ {G, B} (first green, second blue)

$\frac{42}{132} = \frac{21}{66} = \frac{7}{22}$ {G, G} (both are green)

ii) from above calculated probabilities,

$$Pr[\text{Second Blue}] = Pr[BB] + Pr[GB]$$

$$= \frac{5}{33} + \frac{35}{132}$$

$$= \frac{660 + 1155}{4356}$$

$$= \frac{1815}{4356} = \frac{5}{12} \approx 0.41$$

2

iii) from above calculated probability of the balls that are not blue.

$$Pr[\text{At least one blue}] = 1 - Pr[GG]$$

$$= 1 - \frac{7}{22}$$

$$= \frac{22-7}{22} = \frac{15}{22} \approx 0.68$$

2

iv) {B, B} — 2 times , {G, B} — 1 time , {B, G} — 1 time , {G, G} — none.

$$= 2\left(\frac{5}{33}\right) + 1\left(\frac{35}{132}\right) + 1\left(\frac{35}{132}\right) + 0\left(\frac{7}{22}\right)$$

2

$$= \frac{10}{66} + \frac{35}{132} + \frac{35}{132} = \cancel{\frac{1320+35+35}{8712}} = \cancel{\frac{1390}{8712}} = \frac{5}{6} \approx 0.83$$

Original Blue $= \frac{5}{12}$, so repetition will multiply it with a factor of n and make it $\frac{5n}{12}$ ...

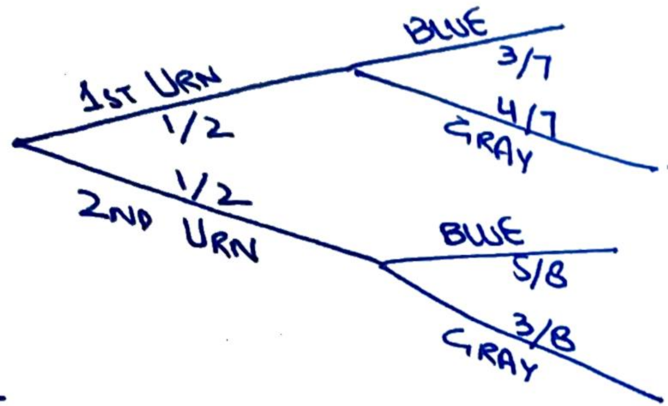
(i think)

$\frac{10}{10}$

QUESTION #2

Urns : 2

1st round — 3 blue, 4 gray 2nd round – 5 blue, 3 gray



$Pr[B_1] = \frac{3}{14}$

$Pr[Gr_1] = \frac{4}{14} = \frac{2}{7}$

$Pr[B_2] = 5/16$

$Pr[Gr_2] = 3/16$

TOTAL BLUE PROBABILITY :

$$Pr[B] = Pr[B_1] + Pr[B_2]$$

$$= \frac{3}{14} + \frac{5}{16}$$

$$= \frac{59}{112}$$

5

Now, to take probability that it is from the first urn.

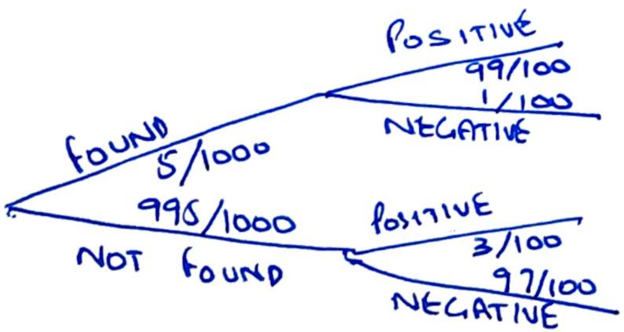
$\frac{Pr[E]}{Pr[S]}$, so.

5

$$Pr[B1] = \frac{Pr[B_1]}{Pr[B]} = \frac{3/14}{59/112} = \frac{24}{59} \approx 0.40$$

10/10

QUESTION # 3



$Pr[FP] = 495/100000$

$Pr[FNg] = 5/100000$

$Pr[NP] = 2985/100000$

$Pr[NNg] = 96515/100000$

$$Pr[P] = Pr[FP] + Pr[NP]$$

$$= 495/100000 + 2985/100000 = 87/2500 \approx 0.03$$

5

$$Pr[F \mid P] = \frac{Pr[FP]}{Pr[P]} = \frac{495/100000}{87/2500} = \frac{33}{232} \approx 0.14$$

5

$$Pr[Ng] = Pr[FNg] + Pr[NNg]$$

$$= 5/100000 + 96515/100000 = \frac{2413}{2500} \approx 0.96$$

$$Pr[N \mid Ng] = \frac{Pr[NNg]}{Pr[N]} = \frac{96515/100000}{2413/2500} = \frac{96515}{96520} \approx 0.99$$

QUESTION #4

5
5

A & B are independent so,

$$Pr[A \cap B] = Pr[A] \times Pr[B] \quad \text{---} (i)$$

To prove,

$$Pr[A \cap B^c] = Pr[A] \times Pr[B^c] \quad \text{---} (ii)$$

Using probability of complement, we can say that

$$Pr[B^c] = 1 - Pr[B] \quad \text{---} (iii)$$

As, $Pr[A \cap B] = Pr[A] \times Pr[B]$ is independent, we can substitute it in (ii), as

$$Pr[A \cap B^c] = Pr[A] - Pr[A] \times Pr[B]$$

$$Pr[A \cap B^c] = Pr[A] \times (1 - Pr[B])$$

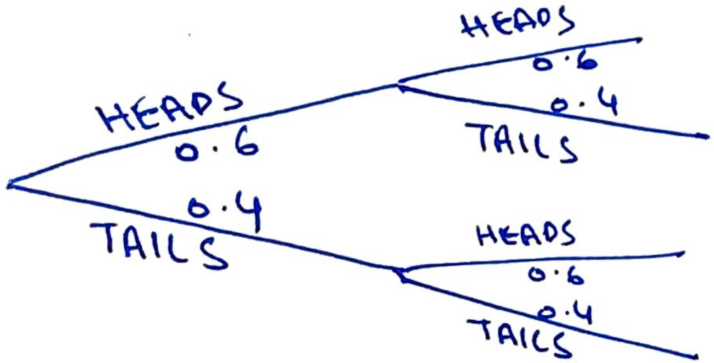
$$Pr[A \cap B^c] = Pr[A] \times Pr[B^c] \quad (\text{from } iii)$$

We have proved that A and B^c are also independent.

QUESTION # 5

10/10

HEADS : 0.6 Tails : 0. ____ ??? HAIN? OK, Q.6 states that



Pr [HH] = 0.36

Pr [HT] = 0.24

Pr [TH] = 0.24

Pr [TT] = 0.16

2.5

i) Pr [HH] = 0.36

ii) Pr [HT] + Pr [TH] = 0.24 + 0.24 = 0.48 2.5

iii) Pr [TT] = 0.16 (No H) 2.5

iv) Pr [HH] + Pr [HT] + Pr [TH] = 0.36 + 0.24 + 0.24 = 0.84 2.5

$\frac{5}{5}$ good :)

QUESTION # 6

HEADS : 0.6 TAILS : 0.4

MUST BE 8 HEADS SO

i) $Pr[H_8T_2] = (0.6)^8 \cdot (0.4)^2 \cdot {}^{10}C_8 = (0.6)^8 \cdot (0.4)^2 \cdot \frac{90}{2}$

$= 0.12$

ii) $Pr[H] = Pr[H_8T_2] + Pr[H_9T_1] + Pr[H_{10}T_0]$

$= 0.12 + {}^{10}C_9 \cdot (0.6)^9 \cdot (0.4) + (0.6)^{10}$

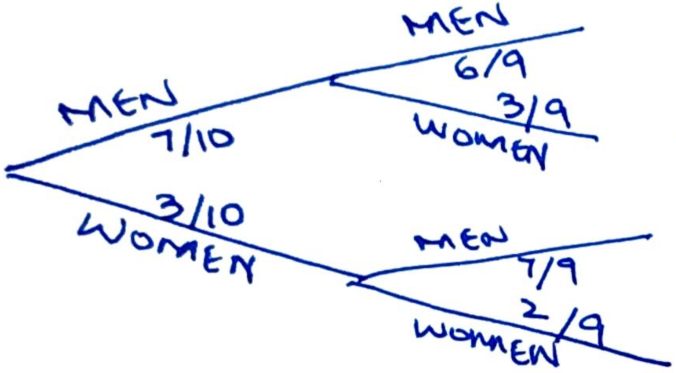
$= 0.12 + (10)(0.010077)(0.4) + (6.04 \ldots \times 10^{-3})$

$= 0.16$

Question No # 7

$\frac{10}{10}$

MEN : 7   WOMEN : 3   Total : 10



$Pr[MM] = \frac{42}{90} = \frac{21}{45} = \frac{7}{15}$

$Pr[MW] = \frac{21}{90} = \frac{7}{30}$

$Pr[WM] = \frac{21}{90} = \frac{7}{30}$

$> 7/15$

$Pr[WW] = \frac{6}{90} = \frac{2}{30} = 1/15$

i) $Pr[WW] = 1/15$

3

ii) $Pr[MM] = 7/15$

3

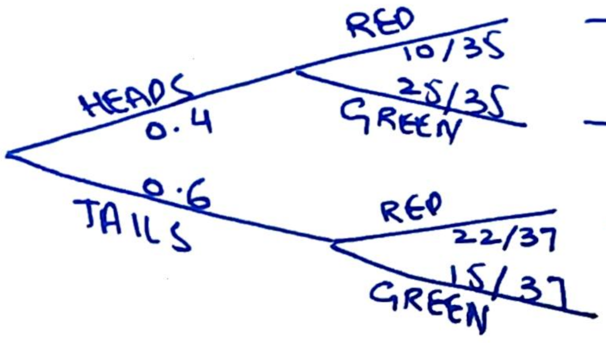
iii) $Pr[MW] + Pr[WM] = 7/30 + 7/30$

$= 7/15$

4

QUESTION # 8

$\frac{10}{10}$



$Pr[HR] = \frac{40}{350} = \frac{4}{35}$

$Pr[HG] = \frac{100}{350} = \frac{10}{35} = \frac{2}{7}$

$Pr[TR] = \frac{132}{370} = \frac{66}{185}$

$Pr[TG] = \frac{90}{370} = \frac{45}{185}$

i) $Pr[G] = Pr[HG] + Pr[TG] = \frac{2}{7} + \frac{45}{185} = \frac{137}{259} \approx 0.52$

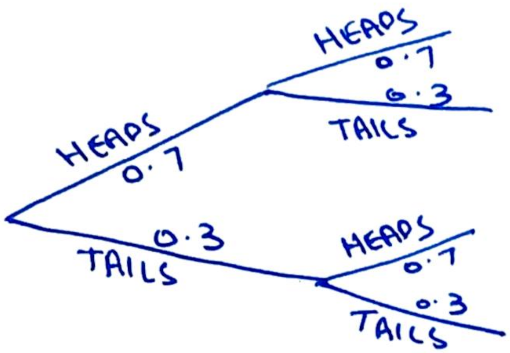
ii) 1ST URN GREEN $= Pr[HG]$

$$Pr[H|G] = \frac{Pr[HG]}{Pr[G]} = \frac{2/7}{137/259} = 74/137 \approx 0.54$$

QUESTION # 9

$\frac{10}{10}$

HEADS : 0.7   Tails : 0.3



Pr[HH] = 0.49

Pr[HT] = 0.21

Pr[TH] = 0.21

Pr[TT] = 0.09

$P(A|B) = \frac{P(A \cap B)}{P(B)}$

i) Pr[HH] = 0.49   2.5

ii) Pr[HT] + Pr[TH] = 0.21 + 0.21 = 0.42   2.5

iii) Pr[TT] = 0.09 (No H)   2.5

iv) Pr[HH] + Pr[HT] + Pr[TH] = 0.49 + 0.21 + 0.21 = 0.91   2.5